

In order for an AI agent to produce an output, machine learning techniques must be used to transform the ingested data into meaningful input. There are two main types of techniques: unsupervised and supervised. Unsupervised techniques involve feeding unlabelled data to an AI algorithm, allowing it to autonomously learn a representation of the data. Supervised techniques, on the other hand, involve feeding labelled data to the algorithm. However, ethical issues arise in both cases if the data contains biases, either from pre-filtering by the data scientist or from the data itself. In such cases, the algorithm will reproduce and even amplify these biases.

Research has shown that AI agents can learn implicit patterns related to sexism, racism, gender inequalities, and other discriminatory behaviors towards minorities. Biases also persist in supervised learning, where the data labelling process is often accompanied by data pre-filtering or selection. This selection may reflect the data curator's own interpretation of reality, resulting in bias towards minorities.

Furthermore, biases and discrimination can be amplified as an AI agent performs feature extraction, as certain features may reinforce societal biases. For example, the use of zip codes in credit risk assessment may lead to discrimination towards certain ethnicities who live in segregated areas. If not properly controlled, AI agents may replicate and reinforce societal fractures and inequalities, transmitting and amplifying prejudices in their output.

As shown by research, algorithms using race as a feature may mistakenly flag black defendants as being at a higher risk of committing future crimes and incorrectly flag white defendants as being at a lower risk. Therefore, it is essential to address and control biases in AI algorithms to ensure fair and ethical outcomes.

Although there are moral debates surrounding generative AI, its future is bright and limitless. The growing trend of reusing IT infrastructure is fueling the generative AI market, reducing barriers to entry and encouraging widespread adoption of the technology. With tools like GitHub Copilot and Builder.ai readily available, indie developers are expected to create exciting new programs and platforms.

Machine learning is no longer exclusive, and more industries than ever can gain a competitive advantage by using AI to optimize workflows, analytics processes, and support programs for customers or employees. Furthermore, according to Gartner, by 2025, at least 30% of new drugs and discovered materials will come from generative AI models.